

DBMS-7-10

```
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Log(message)
    VALUES ('Employee updated');
END //

DELIMITER ;

DELIMITER //

CREATE TRIGGER before_employee_delete
BEFORE DELETE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.emp_id = 1 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee cannot be deleted';
    END IF;
END //

DELIMITER ;

DELIMITER //

CREATE TRIGGER after_employee_delete
AFTER DELETE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Log(message)
    VALUES ('Employee deleted');
END //

DELIMITER ;

INSERT INTO Employee VALUES
(7, 'Rohan', 'HR', -5000);

UPDATE Employee
SET salary = -1000
```

```
WHERE emp_id = 2;
```

```
UPDATE Employee  
SET salary = 60000  
WHERE emp_id = 3;
```

```
DELETE FROM Employee  
WHERE emp_id = 7;
```

```
SELECT * FROM Employee;
```

```
SELECT * FROM Employee_Log;
```

OUTPUT:

emp_id

emp_id	name	department	salary
1	Rahul	IT	50000
2	Priya	HR	40000
3	Aman	IT	60000
4	Neha	HR	45000
5	Karan	Sales	35000

emp_id	name	department	salary
1	Rahul	IT	55000
2	Priya	HR	0
3	Aman	IT	60000
4	Neha	HR	45000
5	Karan	Sales	35000

log_id	message	log_time
1	Employee inserted	2026-10-04 08:51:47
2	Employee updated	2026-10-04 08:51:47
3	Employee updated	2026-10-04 08:51:47
4	Employee deleted	2026-10-04 08:51:47

```
mysql>
mysql> SELECT * FROM Student;
```

Student_ID	Student_Name	Dept_ID	Marks
201	Aditya	11	82
202	Kavya	22	91
203	Rohit	11	78
204	Ishita	33	87
205	Manoj	22	94
206	Divya	55	73

```
6 rows in set (0.00 sec)
```



```
mysql>
mysql> SELECT * FROM Department;
```

Dept_ID	Department	HOD
11	AI & DS	Dr. Kumar
22	Mechanical	Dr. Patel
33	EEE	Dr. Iyer
44	Biotechnology	Dr. Verma

```
4 rows in set (0.00 sec)
```



```
mysql>
mysql>
mysql> -- 1. INNER JOIN
mysql> -- Returns only matching records from both tables.
mysql>
mysql> SELECT
-> Student.Student_ID,
-> Student.Student_Name,
-> Student.Dept_ID,
-> Department.Department,
-> Department.HOD,
-> Student.Marks
-> FROM Student
-> INNER JOIN Department
-> ON Student.Dept_ID = Department.Dept_ID;
```

```

mysql>
mysql>
mysql> -- 3. RIGHT JOIN
mysql> -- Returns all records from the right table and matching records from the left table.
mysql>
mysql> SELECT
->     Student.Student_ID,
->     Student.Student_Name,
->     Student.Dept_ID,
->     Department.Department,
->     Department.HOD,
->     Student.Marks
-> FROM Student
-> RIGHT JOIN Department
-> ON Student.Dept_ID = Department.Dept_ID;

```

Student_ID	Student_Name	Dept_ID	Department	HOD	Marks
203	Rohit	11	AI & DS	Dr. Kumar	78
201	Aditya	11	AI & DS	Dr. Kumar	82
205	Manoj	22	Mechanical	Dr. Patel	94
202	Kavya	22	Mechanical	Dr. Patel	91
204	Ishita	33	EEE	Dr. Iyer	87
NULL	NULL	NULL	Biotechnology	Dr. Verma	NULL

6 rows in set (0.00 sec)

Name	Marks
Aarav	84

1 row in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Department
-> FROM Student
-> WHERE Department IN (
->     SELECT Department
->     FROM Student
->     WHERE City = 'Mysore'
-> );
```

Name	Department
Diya	Mechanical
Karthik	Mechanical

2 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > ALL (
->     SELECT Marks
->     FROM Student
->     WHERE Department = 'EEE'
-> );
```

Name	Marks
Diya	91
Karthik	95

2 rows in set (0.00 sec)

Name	Marks
Diya	91
Ananya	88
Karthik	95
Varun	86

4 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Department
      -> FROM Student
      -> WHERE Department = (
      ->     SELECT Department
      ->     FROM Student
      ->     WHERE Name = 'Aarav'
      -> );
```

Name	Department
Aarav	AI & DS
Rishabh	AI & DS
Ishita	AI & DS

3 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Marks
      -> FROM Student
      -> WHERE Department = 'AI & DS'
      -> AND Marks > (
      ->     SELECT AVG(Marks)
      ->     FROM Student
      ->     WHERE Department = 'AI & DS'
      -> );
```

1 row in set (0.00 sec)

mysql>

mysql>

```
mysql> SELECT Name, Marks
      -> FROM Student
      -> WHERE Marks > (
      ->     SELECT AVG(Marks)
      ->     FROM Student
      -> );
```

Name	Marks
Diya	91
Ananya	88
Karthik	95
Varun	86

4 rows in set (0.00 sec)

mysql>

mysql>

```
mysql> SELECT Name, Marks
      -> FROM Student
      -> WHERE Marks = (
      ->     SELECT MAX(Marks)
      ->     FROM Student
      -> );
```

Name	Marks
Karthik	95

1 row in set (0.00 sec)

mysql>

mysql>

```
mysql> SELECT Name, Marks
      -> FROM Student
      -> WHERE Marks > (
      ->     SELECT Marks
      ->     FROM Student
      ->     WHERE Name = 'Aarav'
      -> );
```

Name	Marks
Aarav	84
Rishabh	77
Ishita	71

3 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Marks
      -> FROM Student
      -> WHERE City = 'Coimbatore'
      -> AND Marks > 80;
```

Name	Marks
Aarav	84

1 row in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT MAX(Marks) AS Highest_Marks
      -> FROM Student;
```

Highest_Marks
95

1 row in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT AVG(Marks) AS Average_Marks
      -> FROM Student;
```

Average_Marks
84.5714

```
mysql>
mysql>
mysql> SELECT Name, Department, Marks
-> FROM Student;
```

Name	Department	Marks
Aarav	AI & DS	84
Diya	Mechanical	91
Rishabh	AI & DS	77
Ananya	EEE	88
Karthik	Mechanical	95
Ishita	AI & DS	71
Varun	EEE	86

7 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > 80;
```

Name	Marks
Aarav	84
Diya	91
Ananya	88
Karthik	95
Varun	86

5 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Department = 'AI & DS';
```

```
mysql> CREATE DATABASE CollegestudentDB;
Query OK, 1 row affected (0.01 sec)
```

```
mysql>
mysql> USE CollegeStudentDB;
Database changed
```

```
mysql>
mysql> CREATE TABLE Student (
-> Student_ID INT PRIMARY KEY,
-> Name VARCHAR(50),
-> Department VARCHAR(20),
-> Marks INT,
-> City VARCHAR(30)
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> INSERT INTO Student VALUES
-> (301, 'Aarav', 'AI & DS', 84, 'Coimbatore'),
-> (302, 'Diya', 'Mechanical', 91, 'Mysore'),
-> (303, 'Rishabh', 'AI & DS', 77, 'Kochi'),
-> (304, 'Ananya', 'EEE', 88, 'Madurai'),
-> (305, 'Karthik', 'Mechanical', 95, 'Pune'),
-> (306, 'Ishita', 'AI & DS', 71, 'Salem'),
-> (307, 'Varun', 'EEE', 86, 'Trichy');
```

Query OK, 7 rows affected (0.01 sec)

Records: 7 Duplicates: 0 Warnings: 0

```
mysql>
mysql>
mysql> SELECT *
-> FROM Student;
```

Student_ID	Name	Department	Marks	City
301	Aarav	AI & DS	84	Coimbatore
302	Diya	Mechanical	91	Mysore
303	Rishabh	AI & DS	77	Kochi
304	Ananya	EEE	88	Madurai
305	Karthik	Mechanical	95	Pune
306	Ishita	AI & DS	71	Salem
307	Varun	EEE	86	Trichy

```

mysql>
mysql>
mysql> -- 4. FULL OUTER JOIN
mysql> -- Returns all records from both tables.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID
    ->
    -> UNION
    ->
    -> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;

```

Student_ID	Student_Name	Dept_ID	Department	HOD	Marks
201	Aditya	11	AI & DS	Dr. Kumar	82
202	Kavya	22	Mechanical	Dr. Patel	91
203	Rohit	11	AI & DS	Dr. Kumar	78
204	Ishita	33	EEE	Dr. Iyer	87
205	Manoj	22	Mechanical	Dr. Patel	94
206	Divya	55	NULL	NULL	73
NULL	NULL	NULL	Biotechnology	Dr. Verma	NULL

7 rows in set (0.01 sec)

```
mysql>
mysql>
mysql> -- 3. RIGHT JOIN
mysql> -- Returns all records from the right table and matching records from the left table.
mysql>
mysql> SELECT
->     Student.Student_ID,
->     Student.Student_Name,
->     Student.Dept_ID,
->     Department.Department,
->     Department.HOD,
->     Student.Marks
-> FROM Student
-> RIGHT JOIN Department
-> ON Student.Dept_ID = Department.Dept_ID;
```

Student_ID	Student_Name	Dept_ID	Department	HOD	Marks
203	Rohit	11	AI & DS	Dr. Kumar	78
201	Aditya	11	AI & DS	Dr. Kumar	82
205	Manoj	22	Mechanical	Dr. Patel	94
202	Kavya	22	Mechanical	Dr. Patel	91
204	Ishita	33	EEE	Dr. Iyer	87
NULL	NULL	NULL	Biotechnology	Dr. Verma	NULL

6 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> -- 2. LEFT JOIN
mysql> -- Returns all records from the left table and matching records from the right table.
mysql>
mysql> SELECT
->     Student.Student_ID,
->     Student.Student_Name,
->     Student.Dept_ID,
->     Department.Department,
->     Department.HOD,
->     Student.Marks
-> FROM Student
-> LEFT JOIN Department
-> ON Student.Dept_ID = Department.Dept_ID;
```

Student_ID	Student_Name	Dept_ID	Department	HOD	Marks
201	Aditya	11	AI & DS	Dr. Kumar	82
202	Kavya	22	Mechanical	Dr. Patel	91
203	Rohit	11	AI & DS	Dr. Kumar	78
204	Ishita	33	EEE	Dr. Iyer	87
205	Manoj	22	Mechanical	Dr. Patel	94
206	Divya	55	NULL	NULL	73

6 rows in set (0.00 sec)